Jaccard Noise-Reduction Validation

Experiment 1 | Semantic similarity vs. semantic similarity + Jaccard | October 2026

Executive finding. Jaccard demonstrably reduces false semantic matches, but the current cutoff also discards many true duplicates. It is a useful noise-filtering signal, **not yet a reliable final duplicate decision rule**.

1. Test setup

200 synthetic decisions in 40 groups of five. We evaluated all within-group pairs (400 total), excluding three ambiguous pairs: **397 labeled comparisons** (120 duplicate, 277 non-duplicate). Results describe pairs, not unique decisions. The dataset is a challenge set, not a production-representative sample.

Measure	Semantic only	Semantic + Jaccard
Correct duplicates (TP)	111 / 120	47 / 120
Incorrect matches (FP)	240	60
Precision	31.6%	43.9%
Recall	92.5%	39.2%
F1 score	47.1%	41.4%

2. What Jaccard contributes

Noise reduction: 180 of 240 false-positive pairs were removed (75%). Precision rose by 12.3 percentage points. This directly supports the hypothesis that lexical overlap can help filter misleading semantic matches.

Cost: 64 genuinely duplicate pairs accepted by semantic similarity were also rejected. Recall fell by 53.3 percentage points, and F1 declined by 5.7 points. At the tested Jaccard cutoff of 0.32, the combined approach does not improve the overall precision-recall balance.

3. Two examples from the experiment

Useful rejection (non-duplicate): "We approved the migration to PostgreSQL." vs. "We rejected the migration to PostgreSQL." Semantic-only accepted; Jaccard rejected. Cosine 0.862; Jaccard 0.714.

Harmful rejection (genuine duplicate): "We approved the migration to PostgreSQL." vs. "The migration to PostgreSQL was approved." Semantic-only accepted; Jaccard rejected. Cosine 0.981; Jaccard 0.714.

These pairs have the same Jaccard score despite opposite ground-truth labels. Word overlap alone cannot distinguish a changed decision from a paraphrase.

4. Conclusion and next steps

Conclusion: Keep Jaccard as a promising noise-reduction signal for investigation, but do not deploy the current strict filter as the final duplicate detector. The next objective is to preserve false-positive reductions while recovering genuine paraphrases.

Recommended evaluation: (1) audit labels and ambiguous cases; (2) test Jaccard thresholds and different ways of using its score, without assuming a solution in advance; (3) analyze errors by action reversal, amount/ID change, and paraphrase; (4) compare on a held-out dataset, reporting precision, recall, F1 and rejected true duplicates; (5)

assess downstream clustering, not just pairs.

Optional research direction - Jev: TypeSafe AI positions Jev as a model for bounded, typed judgments. A separate pilot could ask whether two candidate statements represent the same decision, particularly on Jaccard-rejected true duplicates. This is untested in this experiment and is **not** part of the reported results. Benchmark it against the existing baseline before adoption.

Source: experiment_decision_dataset_200.json; 397-pair experiment results and pair-level examples provided for this validation. Synthetic benchmark; no production claims.